Adele Chinda

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Professional Summary

Ph.D. student in Computer Science at **Georgia State University**, researching **agentic reasoning**, **Large Language Model (LLM) agents**, and **diffusion models**. My research focuses on building intelligent, autonomous systems capable of understanding context, reasoning through uncertainty, and executing complex, multi-step tasks. I am passionate about bridging the gap between high-level cognitive reasoning and low-level robotic control—empowering embodied agents with spatial-semantic memory, adaptive planning, and human-aligned decision-making abilities.

Research and Professional Experience

Ph.D. Researcher *Sep 2025 – Present*

Advanced Mobility & Augmented Intelligence Lab, Georgia State University | Advisor: Dr. Haoxin Wang Atlanta, GA

- Researching multimodal AI systems, diffusion-based language generation, and vision–language reasoning architectures.
- Developing novel mixture-of-experts frameworks for multimodal context-aware understanding.
- Designing collaborative multi-agent systems with semantic reasoning capabilities for complex task execution.
- Investigating the application of diffusion models to enhance LLM performance in controlled text generation and multi-modal understanding.

Ph.D. Researcher *Aug 2024 – Sep 2025*

Assistive Intelligence Lab, Georgia State University | Advisor: Dr. Yi Ding Atlanta, GA

- Investigated how Large Language Models can enhance both high-level and low-level planning in embodied agents.
- Designed reasoning architectures that integrate spatial-semantic memory and goal-conditioned action selection.
- Addressed challenges including partial observability, scaling semantic memory, and occlusion-aware navigation.
- Contributed to the lab's broader mission of developing adaptive, assistive AI systems for real-world collaboration.

Undergraduate Research Assistant *Sep 2023 – Jan 2024*

Artificial Intelligence Lab, Belgorod State University Belgorod, Russia

- Developed intelligent control algorithms for non-linear robotic systems, improving precision and system stability.
- Collaborated with supervisors on data-driven modeling for adaptive robotic performance in uncertain conditions.

ML Research Intern *Jun 2023 – Aug 2023*

CVisionLab Rostov, Russia

- Implemented fuzzy-logic decision systems to model uncertainty in machine perception tasks.
- Conducted experiments comparing fuzzy and probabilistic approaches to image classification and tracking.

Software Engineer *May 2022 – Aug 2023*

Paytrybe Kazan, Russia

- Led end-to-end development of Android and iOS fintech applications, including UI/UX and API integration.
- Established testing and continuous integration workflows to ensure robustness, scalability, and maintainability.

Education

2024–2029 **Ph.D. in Computer Science**, Georgia State University
Advisor: Dr. Haoxin Wang

2020–2024 **B.Sc. in Applied Computer Science**, Belgorod State University, Russia — GPA: 3.89/4.0 (*summa cum laude*)
Supervisor: Assoc. Prof. Evgenia Bolgova

Selected Publications

- **ViMoE: Vision Mixture of Experts with Multimodal Context Awareness**
  World Journal of Advanced Research and Reviews
  A. Chinda. [Project Page] [Code]

- **SEEK-Multi: Collaborative Multi-Agent Semantic Reasoning**
  International Journal of Science and Research Archive
  A. Chinda. [Project Page] [Code]

Selected Research Projects

HierarchicalVLM: Efficient Long-Context Video Understanding via Hierarchical Temporal Aggregation

Project Page — Developed a hierarchical temporal aggregation framework for efficient long-context video understanding using vision–language models.
A. Chinda, D. Emeka, M. Koya, N. N. Ezekwem.

Contrastive Deep Learning for Variant Detection in Wastewater Genomic Sequencing
Project Page *Arxiv* — Built contrastive learning pipelines with VQ-VAE architectures for genomic variant detection in wastewater surveillance data.
A. Chinda*, R. Azumah*, H. D. Venkateswara. **Equal contribution*

GraPO: Graph-based Prediction of Power Outages
Project Page — Developed a machine learning framework leveraging spatiotemporal graph neural networks to predict power outages during extreme weather events.
A. Chinda, H. Chai, J. Morris, J. Fassett.

Predicting Hospital Readmission Risk from EHR Data
Project Page — Designed experiments comparing classical ML and ensemble methods for healthcare outcome prediction, identifying key features driving early patient readmission.
A. Chinda, O. Diallo, Y. Mumin.

Knowledge Graph Analysis of Far-Right Networks
Project Page — Constructed and analyzed large-scale knowledge graphs to uncover online extremist networks and study their information diffusion and community structures.
T. Hossain, **A. Chinda**.

Technical Skills

- **Multimodal AI & Vision–Language:** Vision Transformers (ViT), CLIP, Mixture of Experts (MoE), cross-modal attention, visual grounding, video understanding.
- **Diffusion Models:** Denoising Diffusion Probabilistic Models (DDPM), latent diffusion, score-based generative models, classifier-free guidance, diffusion for text generation.
- **Deep Learning & NLP:** Transformer architectures, large language models (GPT, LLaMA, Mistral), RLHF, LoRA/QLoRA fine-tuning, prompt engineering, tokenization.
- **Reinforcement Learning:** MDPs, deep RL (PPO, SAC, DQN), imitation learning, reward modeling, PDDL planning.

- **Multi-Agent Systems:** Collaborative reasoning, semantic communication, tool-augmented LLM agents, chain-of-thought prompting.
- **Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers & Diffusers, scikit-learn, DeepSpeed, PEFT, Weights & Biases.
- **Simulation & Embodied AI:** AI2-THOR, Habitat, CARLA, Gazebo.
- **Infrastructure & Tools:** Git, Docker, Linux, SLURM, CUDA, distributed training (DDP, FSDP), LaTeX, Jupyter.
- **Programming Languages:** Python, C++, Java, Kotlin, SQL, Bash.
- **Research Competencies:** Experiment design, ablation studies, statistical analysis, scientific writing, peer review, academic presentation.

Affiliations

Advanced Mobility & Augmented Intelligence Lab (AMAI)

Collaborative Human-AI Center (CHAI)

ML Collective

Black in AI

National Society of Black Engineers (NSBE)

Association for Computing Machinery (ACM)